

Understanding Different Height Systems in Geodesy

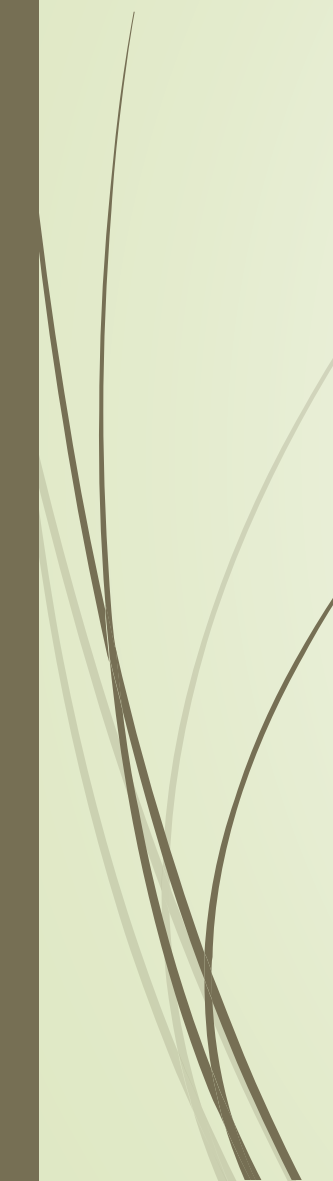


WHAT YOU WILL LEARN TODAY

- The Hidden Force Behind Heights — **Gravity Potential in Motion**
(*gravity plays the most important part in the definition of height, much as it plays the most important role in the definition of weight.*)
- Key height types used in geodesy (h , H , H^N , H^d)
- Measurement techniques: GNSS, leveling, gravity models.
- Importance in applications: mapping, aviation, surveying, hydrology and fluid studies.
- How to convert it between different height systems.

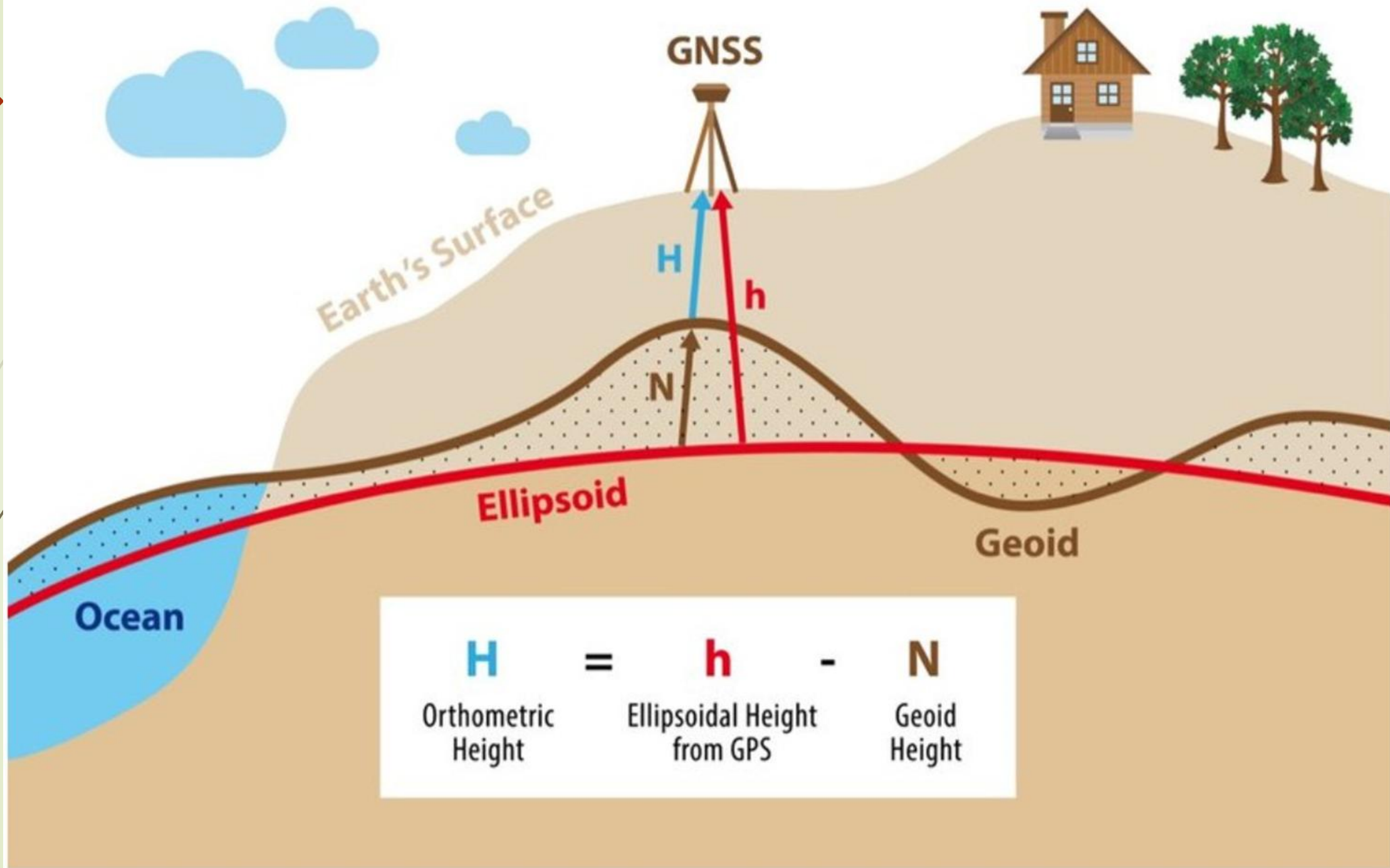


WHY ARE THERE SO MANY HEIGHT SYSTEMS

- The Earth is not a perfect sphere or ellipsoid
 - GNSS provides ellipsoidal height, but we live on the physical surface
 - Mean sea level, gravity, and terrain influence height systems.
 - Real-world applications require different definitions of 'height'.
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MAJOR TYPES OF HEIGHTS IN GEODESY

- **Ellipsoidal Height (h):** From GNSS, relative to ellipsoidal Datum
- **Orthometric Height (H):** From leveling, relative to geoid
- **Normal Height (H^*):** Based on quasigeoid,
- **Dynamic Height (H_d):** Used in hydrology and fluid studies



What Influences the Vertical? A Look at Gravity Components

An observer on the surface of the Earth experiences the combination of the gravitational and centrifugal accelerations. The resultant is the acceleration due to **gravity (g)**

1. Gravitation (b)

- This is the pure attraction of Earth's mass.
- It follows Newton's law of universal gravitation.
- Points toward Earth's center of mass.

2. Centrifugal Acceleration (z)

- Comes from Earth's rotation.
- Acts outward, perpendicular to Earth's rotation axis.
- Its magnitude increases with distance from the axis i.e., strongest at the equator, zero at the poles. Together, They Form:

3. Gravity $g = b + z$

- This is the actual acceleration we measure using gravimeters.
- It defines the direction of the vertical(plumb lines) at any point on Earth.
- The vector sum is tilted slightly away from the Earth's center (because of rotation).

Physical Meaning of This Equation:

Gravity = Gravitational Attraction + Centrifugal Repulsion

Why a plumb line doesn't point exactly toward the center of the Earth?

What Influences the Vertical? A Look at Gravity Components

Consider the effect on the gravity potential of moving a small distance, ds , in any direction. The corresponding change in potential, dW , is given by the *differential*:

$$dW = \frac{\partial W}{\partial x} dx + \frac{\partial W}{\partial y} dy + \frac{\partial W}{\partial z} dz$$

Now, this can be rewritten in vector form as:

$$\begin{aligned} dW &= \left(\frac{\partial W}{\partial x} \mathbf{i} + \frac{\partial W}{\partial y} \mathbf{j} + \frac{\partial W}{\partial z} \mathbf{k} \right) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) \\ &= \nabla W \cdot d\mathbf{s} \end{aligned}$$

Gradient of gravity potential multiply with displacement

So since $\mathbf{g} = \nabla W$ from Eq. (2.17), we have:

$$dW = \mathbf{g} \cdot d\mathbf{s}$$

This dot product can be written as:

$$dW = |\mathbf{g}| |d\mathbf{s}| \cos \theta$$

where θ is the angle between the vectors $\mathbf{g}$ and $d\mathbf{s}$.

Gravity as Gradient of Potential

Let $\mathbf{n}$ be the unit vector normal to an equipotential surface. This is in the zenith direction.

If we move a small distance $d\mathbf{n}$ upwards along $\mathbf{n}$, then,

$$dW = -|\mathbf{g}| |d\mathbf{n}|$$

because $\theta = 180^\circ$ (we are moving against the direction of gravity).

Now, orthometric height (H) is measured along the plumbline, i.e., in the direction of each $d\mathbf{n}$.

$$g = -\frac{\partial W}{\partial H}$$

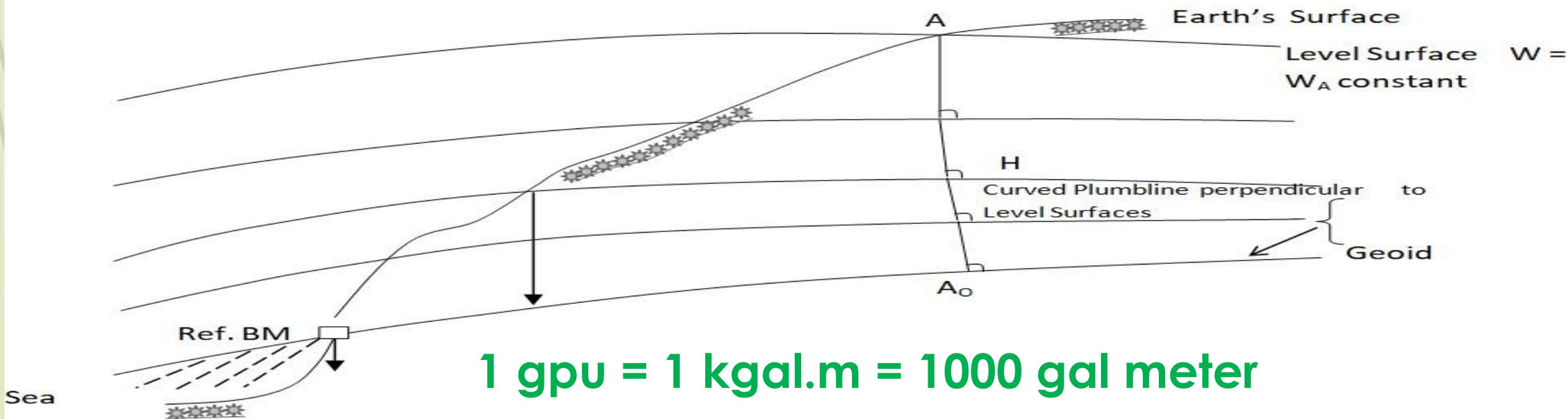
Gravity, hence, is the -ive vertical gradient of potential.

Geopotential numbers C_A are defined by the following equation which gives the change in gravity potential energy between a point on the geoid and another point of interest.

$$C_A = W_0 - W_A = - \int_{A_0}^A dw = \int_{A_0}^A g dh$$

The geopotential number for any place is the potential of the geoid W_0 minus the potential of that place W_A .

C_A is called the **gpn**, which can be determined by observed geometrical height and observed Surface gravity.



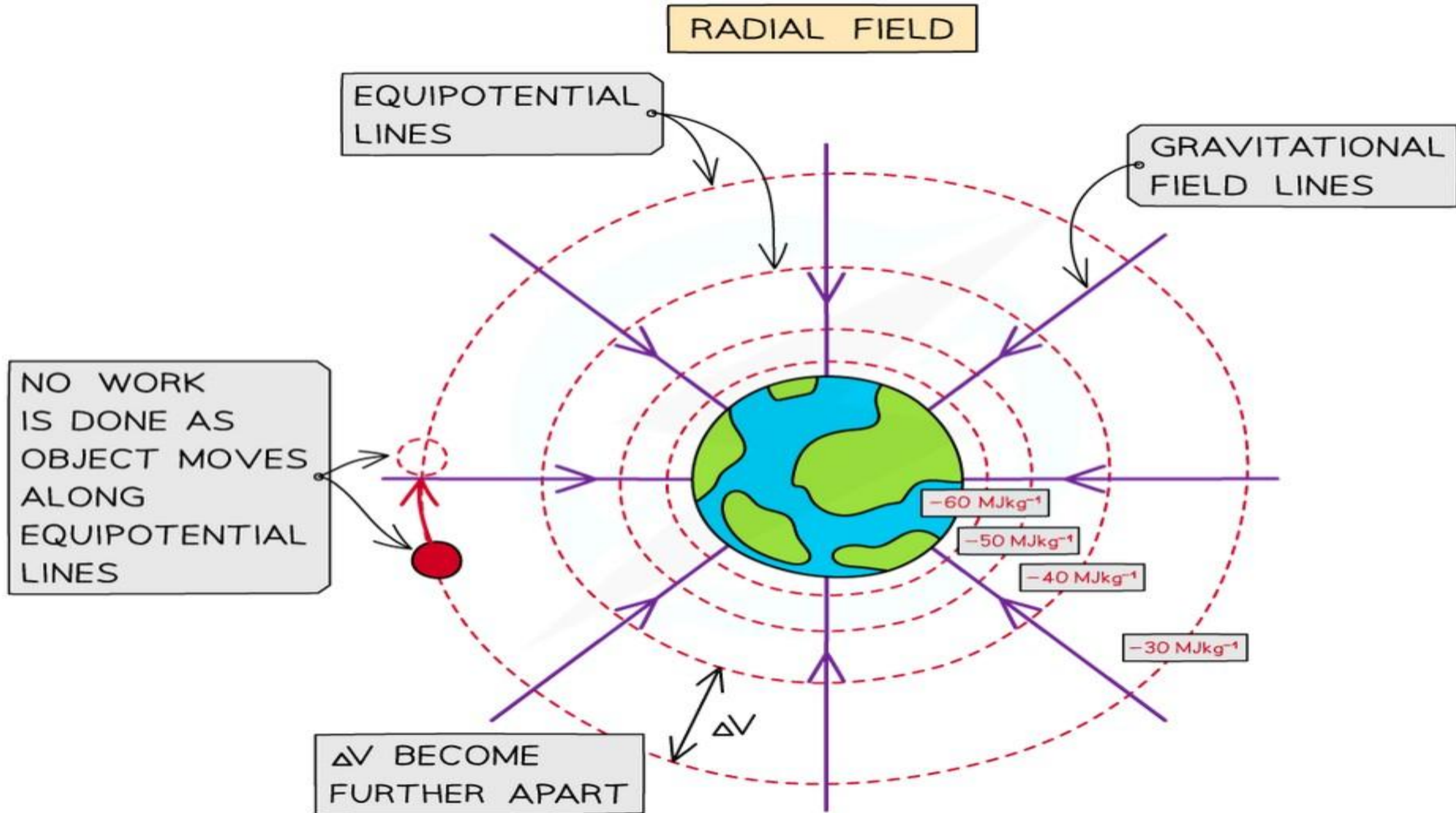
RADIAL FIELD

EQUIPOTENTIAL
LINES

GRAVITATIONAL
FIELD LINES

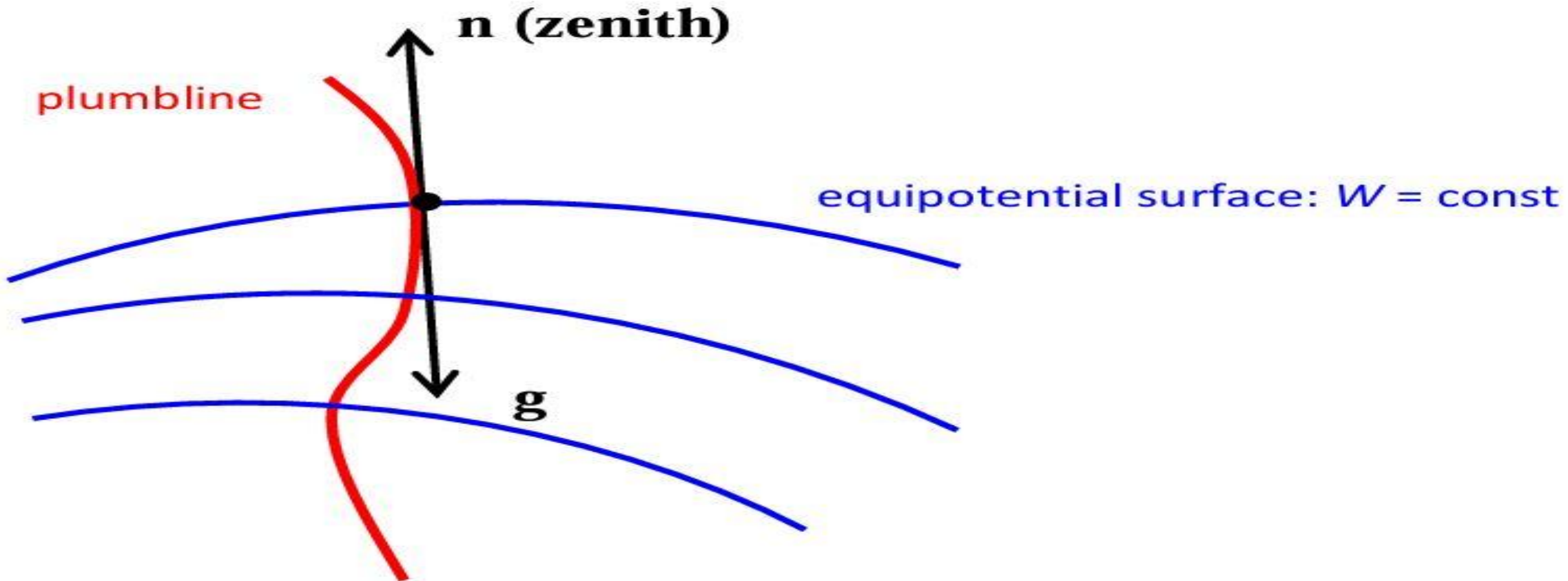
NO WORK
IS DONE AS
OBJECT MOVES
ALONG
EQUIPOTENTIAL
LINES

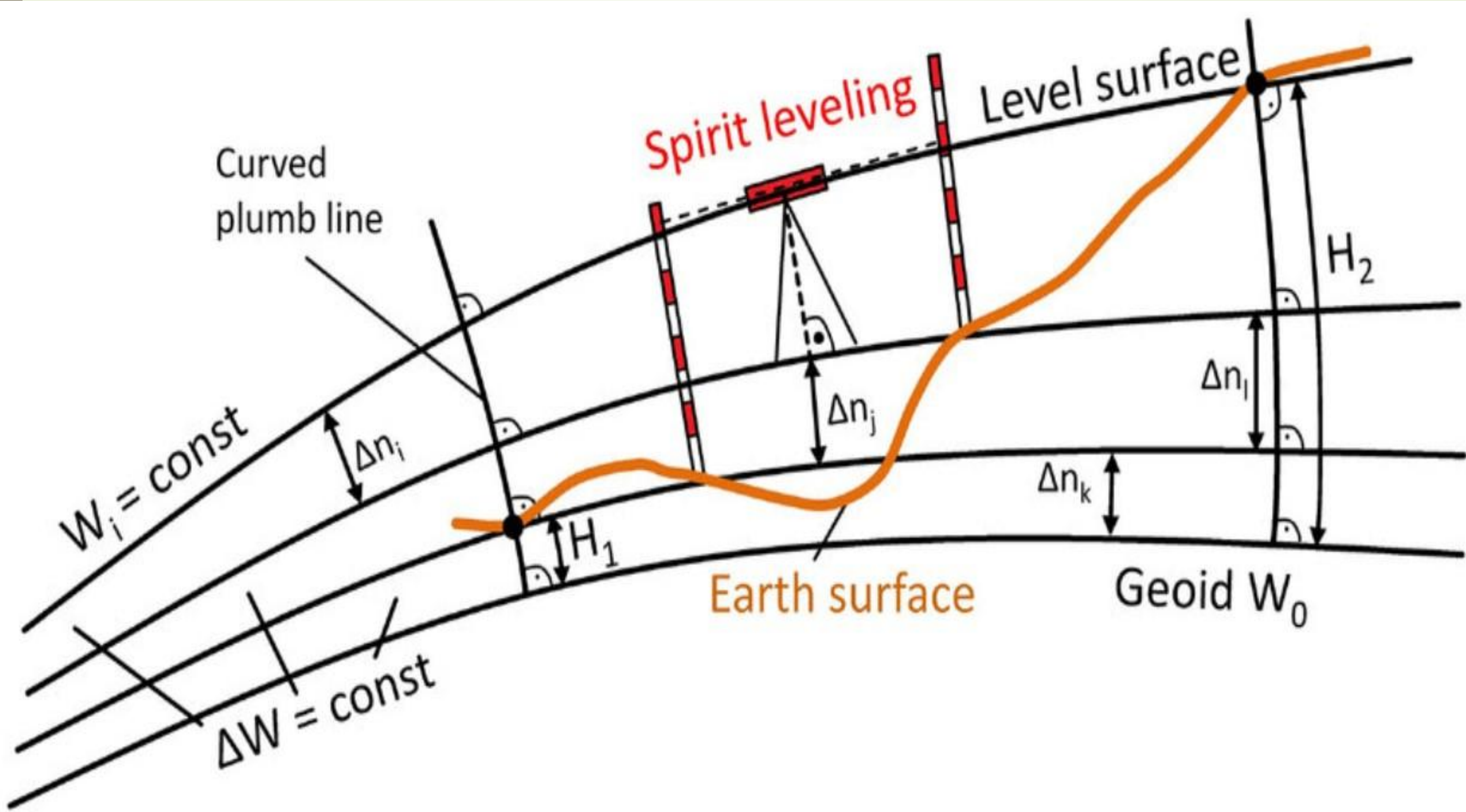
ΔV BECOME
FURTHER APART



EQUIPOTENTIAL SURFACES

gravity vector defines the local vertical or plumbline (setting up an instrument using a spirit bubble), and a surface of constant gravity potential defines the local horizontal





GEOPOTENTIAL NUMBER IS A NATURAL MEASURE OF PHYSICAL HEIGHT

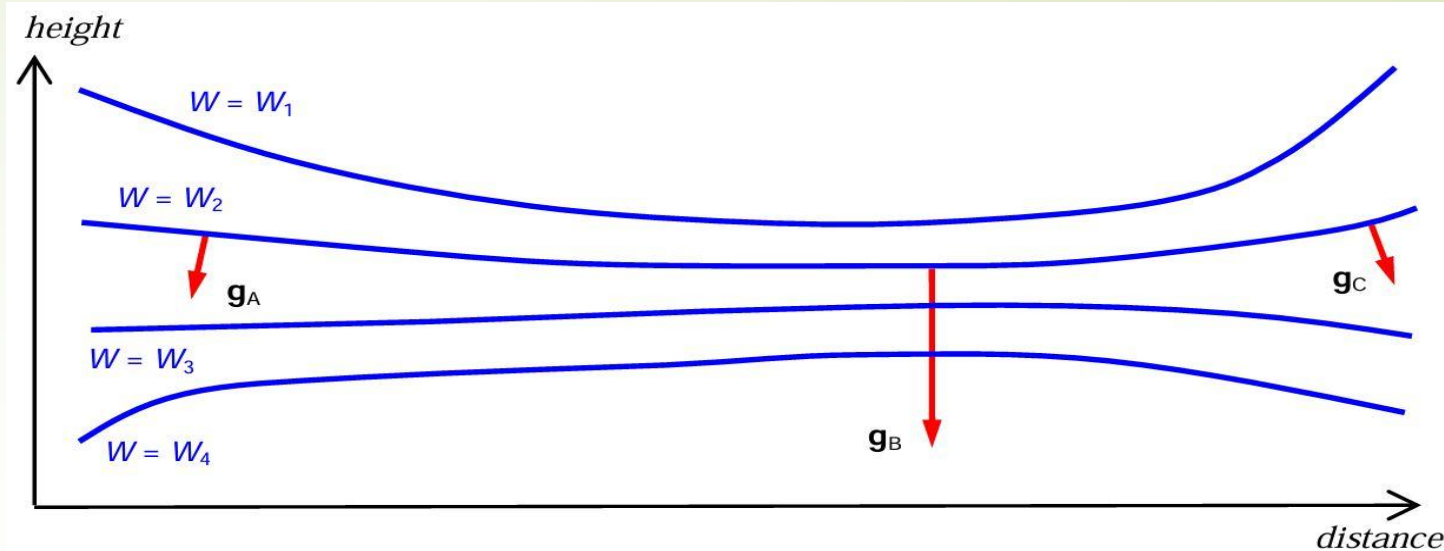
a **gpn** is the potential of the earth's gravity field at any point in space. if two points are at the same **gpn**, water will not flow between them due to the gravity alone.

Geopotential decrease linearly with distance from the centre of the earth. It means the potential decreases with the distance away from the earth, so this difference is the positive number. This makes it a natural measure of distance.

It does not depend on the path taken from the earth's centre to the point of interest.

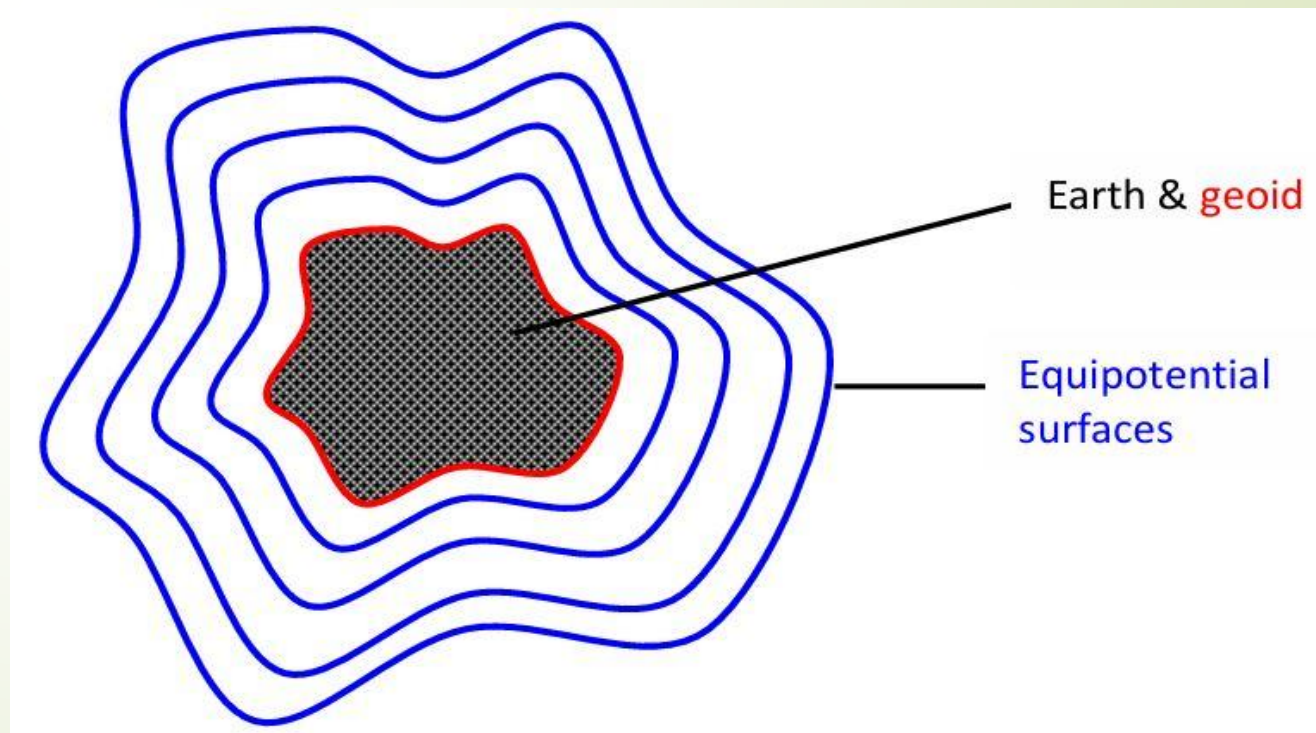
The magnitude of a **gpn** is less important than the relative values between two places. Therefore, we can define the geoid to have a **gpn** zero.

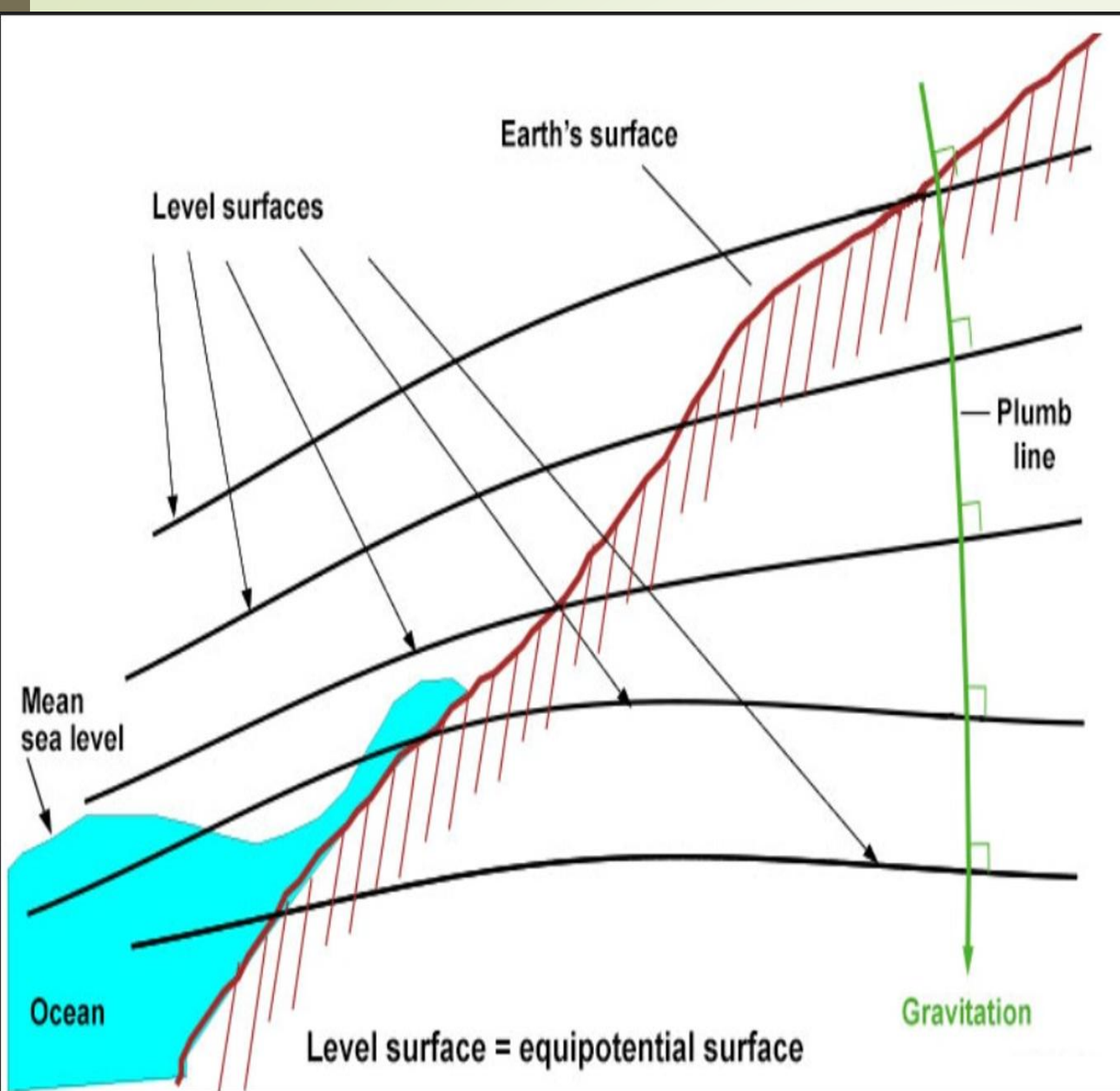
gravity vector points
in the direction of
max. change in the
gravity potential



Equipotential surfaces

The geoid is the one equipotential surface which corresponds most closely with mean sea level





Gravity on an Equipotential Surface

Gravity is not constant on an equipotential surface!!

where the level surfaces are closer together, the gradient is steeper, thus the acceleration is larger;

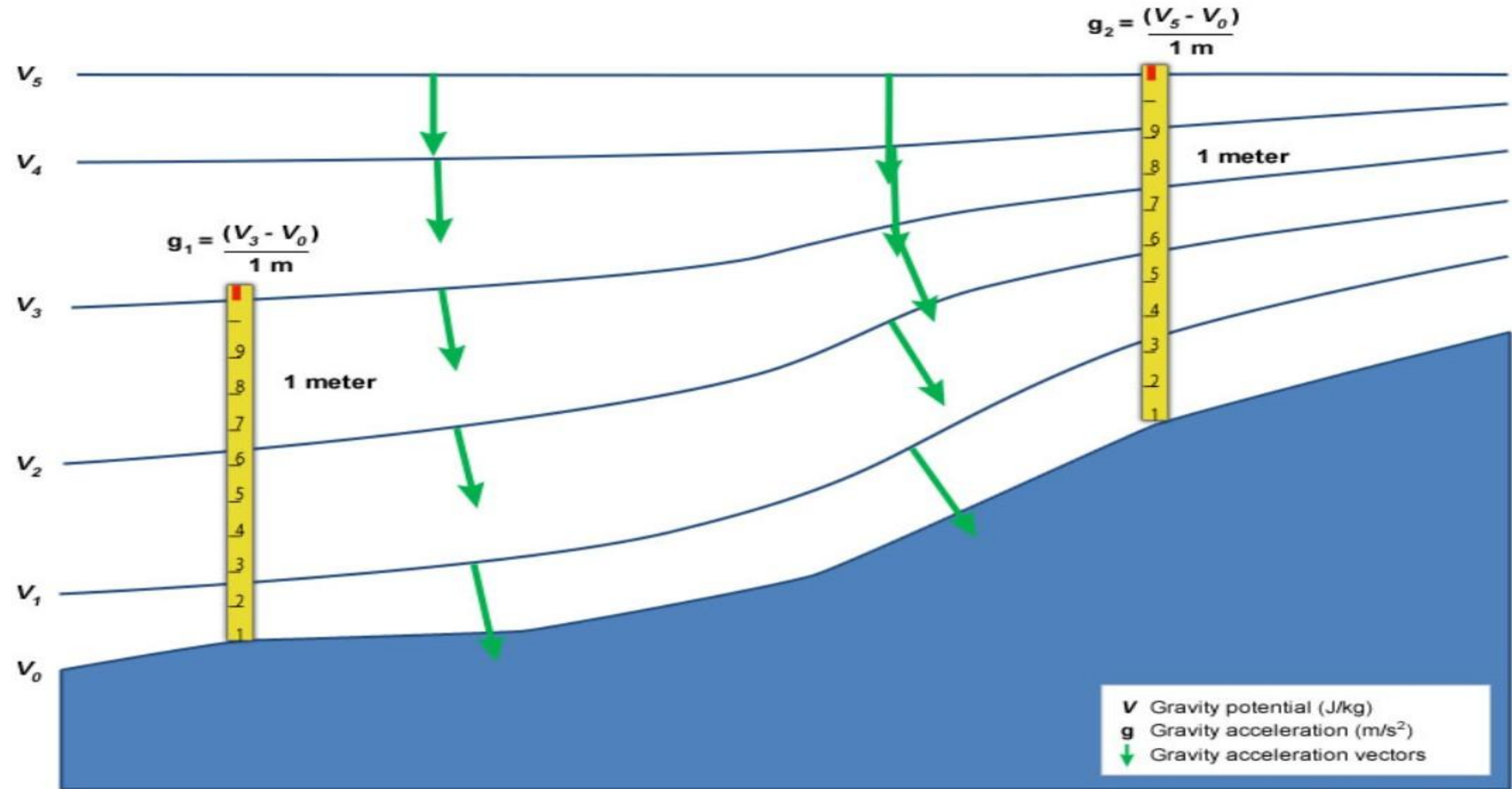
the equipotential surfaces are not parallel in the purely geometrical sense

the gravity vector at the surface of the Earth does not necessarily point towards the geocentre.

equipotential surfaces are always perpendicular to the direction of the plumbline.

When moving along an equipotential surface, gravity potential does not change: $dW = 0$.

Gravity Variations Along Equipotential Surfaces



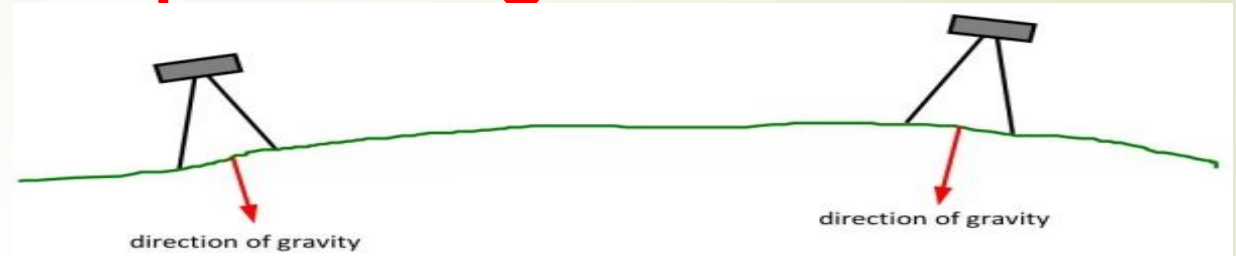
There are a few reasons why geodetic surveyors need to take into account the Earth's gravity field when undertaking a survey and processing its data.

Instrument Set-Up

An instrument set up level in one region will have a different orientation to that in another because the direction of gravity is different at each location.

Misclosures

even if the measurements were all error-free, a closed levelling loop will not necessarily close because the instrument has been aligned differently during each set-up.

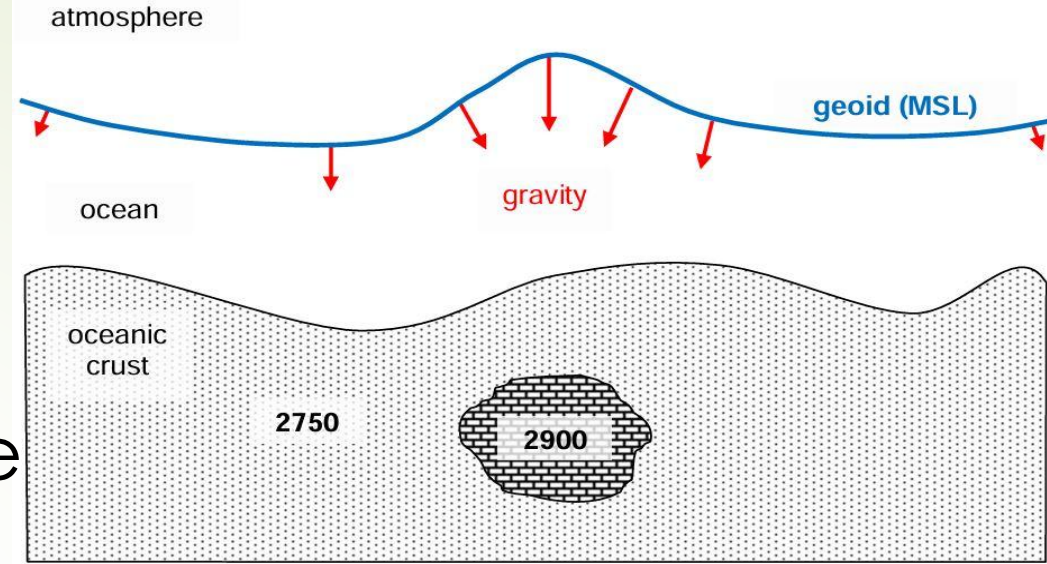


Geoid Computation

Showing how high density rock (2900 kg/m³) emplaced within lower-density seafloor (2750 kg/m³) causes the geoid to rise up..

Since the gravity vector (red arrows) is always perpendicular to an equipotential surface, the direction and magnitude of gravity varies from place to place

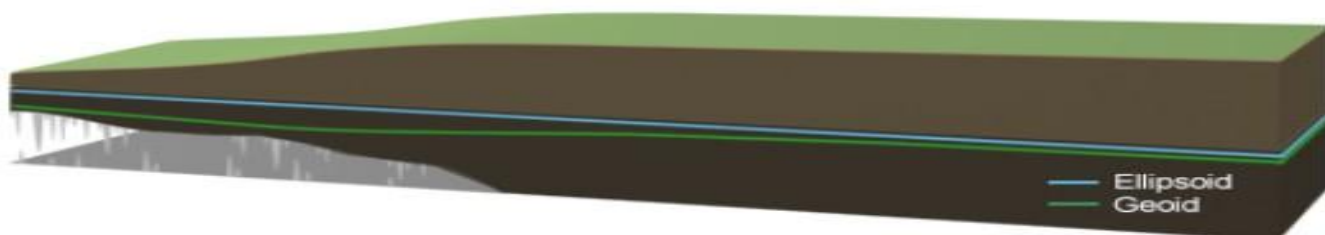
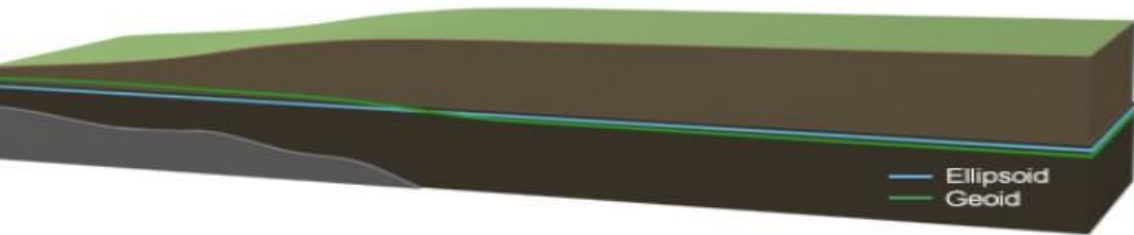
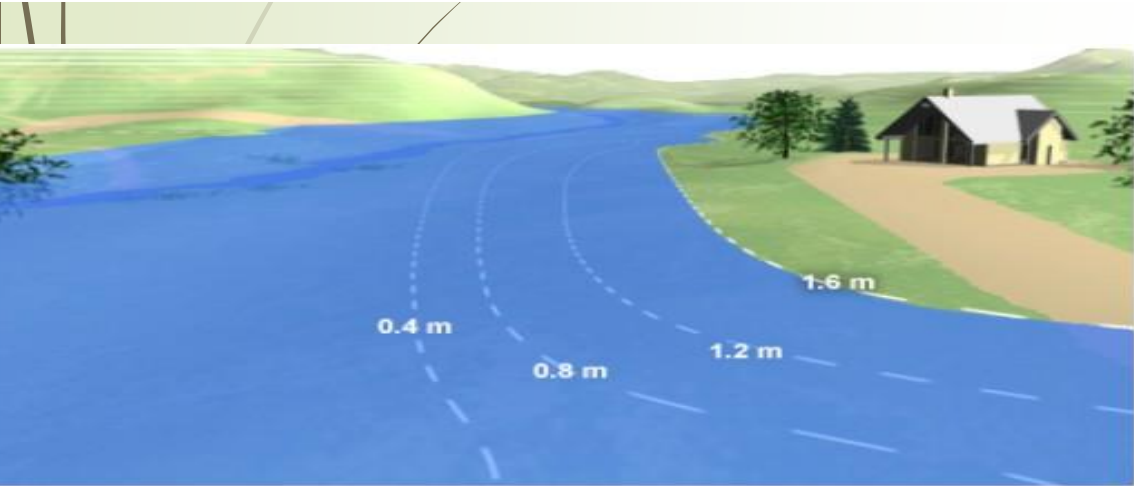
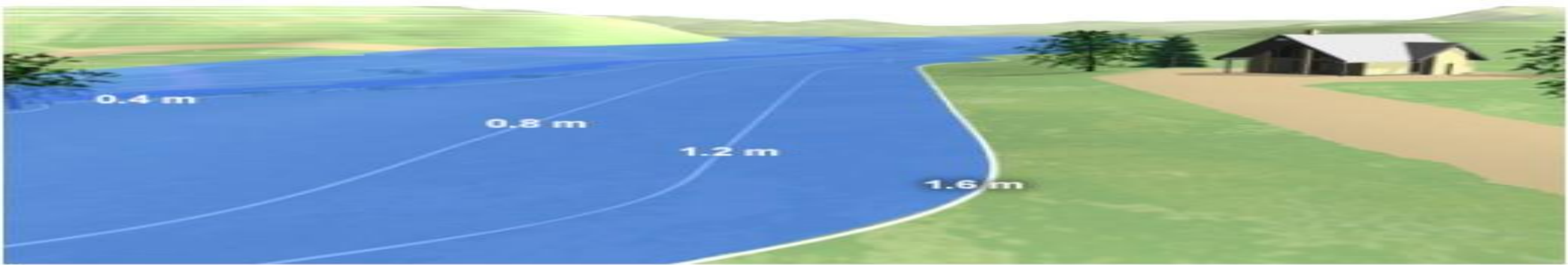
The geoid is the reference surface for physically heights, Shape of the geoid is determined by the earth's gravity potential. geoid is to derive it from gravity measurements, because the geoid height is a function of gravity potential, which can be obtained from measurements of the acceleration due to gravity.



Geodetic Applications that Rely on Measuring Gravity Mapping Water Flow: Small Scale



We know that heights based on the gravity equipotential surface can differ significantly from the geometric heights, due to gravity difference induced by subsurface geology density variations.



Mean gravity $\bar{g}$ along the plumb line

$$\bar{g} = \frac{1}{H} \int_0^H g(z) dz$$

We want to compute physical height from the geoid (mean sea level) to a point along the curved path of gravity — the plumb line. But gravity is not constant with height — it changes (decreases) as we go up. So, we cannot just use one value of gravity — we must use an average of gravity over the height

“Take the gravity at every point z from **0** to **H**, add them up (using the integral), and divide by the total height H to get the mean gravity $\bar{g}$ ”

$$\bar{g} * H = \int_0^H g(z) dz = W_0 - W_P = C_P$$

H Physical height (height above geoid)

C_P Geopotential number at point P ($W_0 - W_P$)

$g(z)$ Gravity along the plumb line (from geoid upward)

$\bar{g}$ Mean gravity between geoid and point P

HELMERT ORTHOMETRIC HEIGHT

Bridging gravity variations and elevation by accounting for internal mass distribution of Earth

What is Orthometric Height?

Orthometric height is the "true elevation above sea level", taking into account Earth's gravity field. Height of a point measured along the Plumb line from the geoid to that point on the Earth's surface. The geoid is the surface of equal gravitational potential that approximates mean sea level.

Why Helmert Height?

- True orthometric height requires knowing gravity **inside the Earth** (impractical).
- It approximates gravity along the plumb line using Helmert's method. (**Helmert's method** uses a mathematical approximation to model internal gravity behavior using **measured surface gravity**.)
- It refines the basic orthometric height formula:

Application:

- Used in **National leveling networks** (e.g., INDIA, USA).
- Ensures height values correspond to **equal potential surfaces**, critical in engineering and hydrology.

"Helmert's approach balances theoretical rigor with practical feasibility — making it a widely adopted orthometric height standard."

HELMERT APPROXIMATION OF ORTHOMETRIC HEIGHT

- For Helmert orthometric heights, gravity measured on the Earth's surface, $g(H)$, is extrapolated downwards by using the estimated vertical gravity gradient interior to the rock.
- 1. It is assumed that its standard value outside the rock, the value -0.3086 mGal/m (the free-air gradient),
- 2. changes to a value that is 0.2238 mGal / m greater (twice the standard-density 2670 kg/m^3 Bouguer plate effect):
- the end result is the total inside-rock gravity gradient, -0.0846 mGal/m

HELMERT APPROXIMATION ORTHOMETRIC HEIGHT

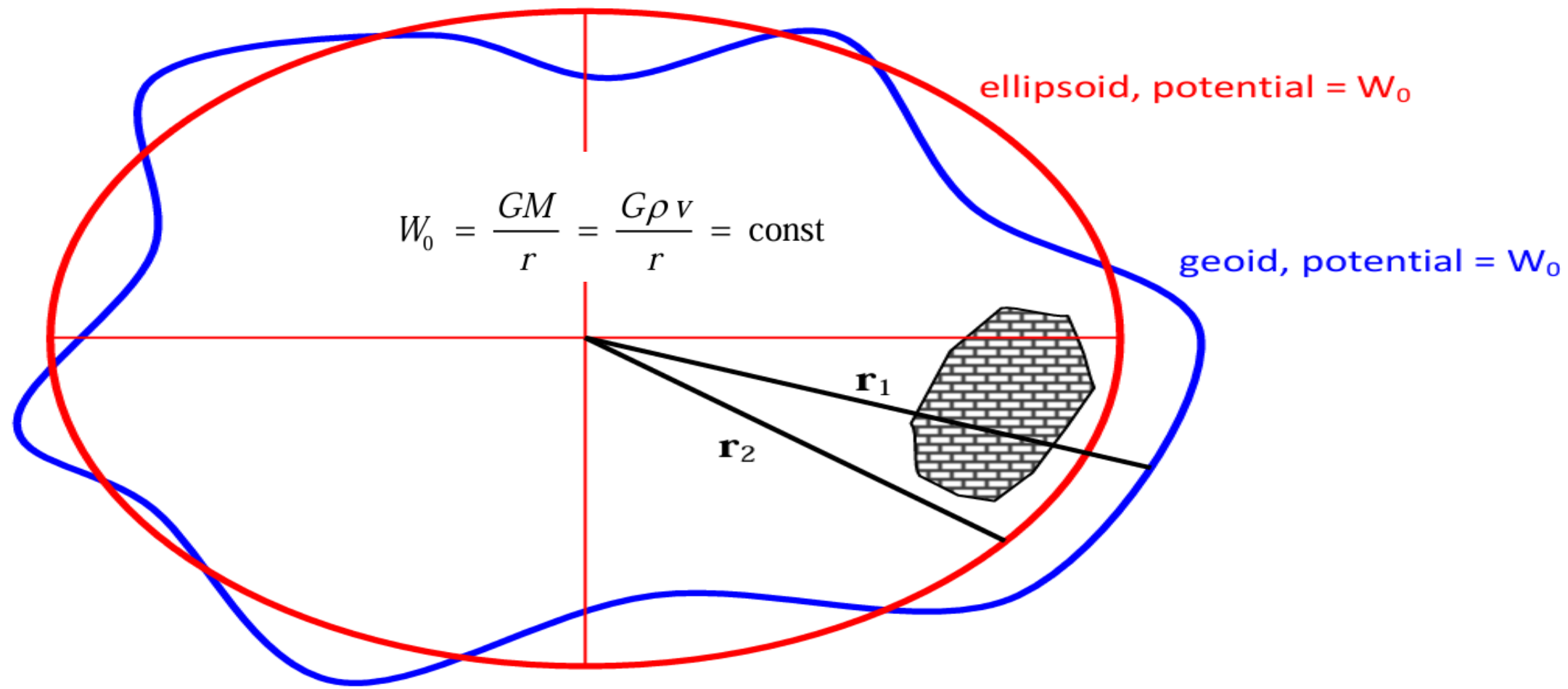
Prey reduction

The coefficient (-0.0848 mGal/m) is half the gravity gradient, so the **mean gravity** along the plumb line is the same as gravity at the **midpoint** of the plumb line

$$\bar{g} = g(H) - 0.0848 \text{ mGal/m} \left(-\frac{1}{2}H\right) = g(H) + 0.0424 \text{ mGal/m} \cdot H$$

$$H = \frac{C}{\bar{g}} = \frac{C}{g(H) + 0.0424 \text{ mGal/m} \cdot H}$$

C is the adjusted geopotential number (potential difference with geoid) and $g(H)$ is gravity at the Earth's surface



In order to calculate the magnitude of the normal gravity on the surface of the normal ellipsoid, the ***Somigliana-Pizzetti formula*** is used:

$$\gamma = \frac{a\gamma_a \cos^2 \phi + b\gamma_b \sin^2 \phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}} \quad (3.8)$$

NORMAL HEIGHT

A gravity-based height system using average normal gravity along the plumb line on a reference ellipsoid

What is Normal Height?

- A height system based on **geopotential numbers** (C) divided by **mean normal gravity** along the plumb line.
- $H^* = C / \bar{\gamma}$
- $C = W_0 - W$ is the **geopotential number**
- $\bar{\gamma}$ is the **mean normal gravity** along the plumb line (approximated using normal gravity field)

Why Use Normal Height?

- Does not require knowledge of actual gravity inside the Earth
- Practical to compute using **surface gravity** and **GRS80 reference ellipsoid**
- Used in many national height systems (e.g., Eastern Europe, Russia)

Normal heights do **not** refer to the geoid, but to the **quasigeoid**, a surface close to the geoid but derived from the **normal gravity field**.

NORMAL HEIGHT

$$H^* = \frac{C}{\bar{\gamma}} = -\frac{W - W_0}{\bar{\gamma}}, \quad (7.6)$$

where the average value of normal gravity along the plumb line is

$$\bar{\gamma} = \overline{\gamma_{0H}} = \frac{1}{H^*} \int_0^{H^*} \gamma(z) dz.$$

Because normal gravity is in good approximation a linear function of z , we may write

$$\bar{\gamma} = \gamma_0 + \frac{1}{2} H^* \frac{\partial \gamma}{\partial z},$$

in which $\frac{\partial}{\partial z} \gamma = -0.3084 \text{ mGal/m}$ and $\gamma_0(\varphi) \stackrel{\text{def}}{=} \gamma(\varphi, 0)$ is normal gravity computed at height zero. We obtain

$$\bar{\gamma} = \gamma_0 - 0.1542 \text{ mGal/m} \cdot H^*.$$

The solution is again obtained iteratively:

$$H^* = \frac{C}{\bar{\gamma}} = \frac{C}{\gamma_0 - 0.1542 \text{ mGal/m} \cdot H^*} \quad (7.14)$$

Dynamic Height:

The geopotential numbers when divided by a specific gravity value, which will change its unit from Kgal.metre to metre and the result so obtained, will be the **Dynamic Height**.

$$H^{\text{dyn}} = C_p / \gamma_0$$

Where γ_0 is the value of normal gravity at latitude 24° which is taken in first level network for India. Normal gravity is determined from a mathematical model approximating actual gravity. The dynamic height has no geometric meaning as it represent a scaled difference of potential. Two different places on the same equipotential surface have the same dynamic height but generally do not have the same orthometric height. Thus dynamic heights are not “distance from the Geoid”.

